WeGrow
Skill Campus & B School



15-Day MERN Stack Internship

Brief Agenda | Session Allocation | Team Projects

15	10	2	6 hrs
Days	Students	Teams	Per Day

1 Session & Trainer Allocation

Session	Handled by	Focus
Morning session	Rajkumar	Primary morning trainer; handles the major morning learning session.
Afternoon session	Lakshmi Priya & Subhashini	Afternoon training, guided practice, coding support and student progress review.

Suggested daily split: 3 hours morning + 3 hours afternoon (within the 6-hour training day). Adjust break timing to the institute timetable.

2 15-Day Agenda – Topic Sequence

Day	Topic	Brief coverage
Day 1	Git & GitHub – Introduction	Version control basics, Git installation, repository, working tree, staging and first commit.
Day 2	Git & GitHub – Practical Workflow	GitHub account/repository, push, pull, clone, branch, merge, README and team collaboration.
Day 3	React – Introduction & Setup	MERN overview, React basics, Vite setup, JSX, components and project structure.
Day 4	React – Components & Data	Props, state, events, conditional rendering, lists and reusable components.
Day 5	React – Forms & Hooks	Controlled forms, validation, useState, useEffect and component lifecycle concepts.
Day 6	React – API Integration	Fetch/Axios basics, JSON, loading/error states, CORS concept and frontend practice.
Day 7	Node.js – Fundamentals	Node.js runtime, npm, package.json, modules, built-in modules and environment variables.
Day 8	Node.js – Asynchronous Programming	Callbacks overview, Promises, async/await, try/catch and a basic Node HTTP server.

Day	Topic	Brief coverage
Day 9	Express.js – Introduction	Express setup, server creation, routes, request/response, HTTP methods and status codes.
Day 10	Express.js – REST API Development	Middleware, route parameters, query strings, request body, controllers and CRUD API testing with Postman.
Day 11	MongoDB – Fundamentals	Database, collections, documents, MongoDB Compass/Atlas and insert, find, update and delete operations.
Day 12	MongoDB + Mongoose Integration	Connect Express to MongoDB, schemas, models, validation and CRUD operations using Mongoose.
Day 13	Team Project – Planning & Backend	Split into two teams of five; finalize requirements, database design, API endpoints and start backend implementation.
Day 14	Team Project – Frontend & Integration	Build React screens, connect APIs, complete CRUD flow, validate forms and fix integration issues.
Day 15	Project Completion & Demo	Testing, bug fixing, GitHub push, README, team presentation, individual explanation and feedback.

3 Team Project Allocation

Divide the 10 students into two teams, with 5 students in each team.

Team A	**Student & Course Management** Student/course CRUD, assign students to courses, search and form validation.
Team B	**Stock / Inventory Management** Product CRUD, stock-in, stock-out, current stock calculation and low-stock indicator.

4 Expected Outcome

- Students understand the workflow from Git/GitHub to frontend, backend and database integration.
- Each team completes one simple working MERN project with CRUD functionality.
- Every student contributes code, commits work to GitHub and explains their contribution.
- Final day includes project demonstration, individual viva and feedback.

5 Recommended Daily 6-Hour Structure

Session	Suggested use
Morning – 3 hours (Rajkumar)	Concept introduction, trainer demonstration and core topic explanation.
Afternoon – 3 hours (Lakshmi Priya & Subhashini)	Hands-on practice, guided coding, exercises, project support and daily review.

Prepared by WeGrow Skill Campus & B School for a 10-student, 15-day practical MERN Stack internship.